

Product Carbon Footprint Study Report

Product Name: liquid crystal display production, unmounted, mobile device

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: liquid crystal display production, unmounted, mobile device

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg liquid crystal display production, unmounted, mobile device

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg liquid crystal display production, unmounted, mobile device.

2. System Boundary

liquid crystal display production, unmounted, mobile device, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of liquid crystal display production, unmounted, mobile device produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **107.096 kg CO2-Eq**.

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

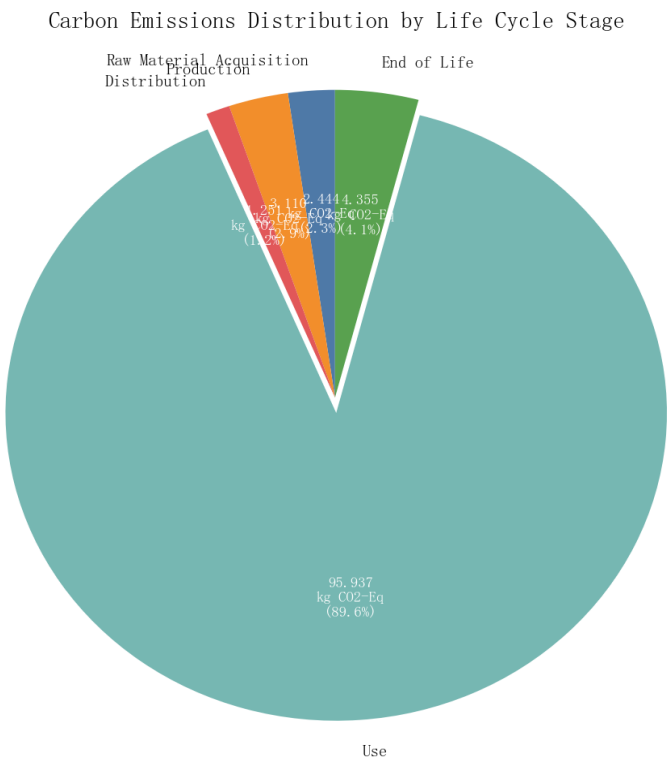


Table 1: Carbon Emissions by Life Cycle Stage

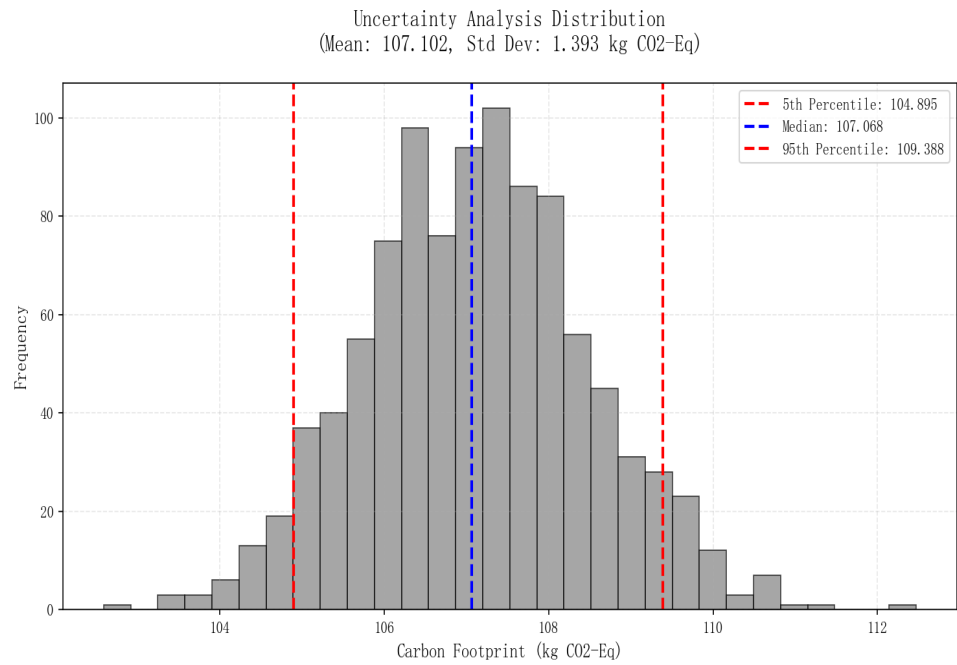
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	2.444	2.28
Production	3.110	2.90
Distribution	1.251	1.17

Use	95.937	89.58
End of Life	4.355	4.07
Total	107.096	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	107.102
Standard Deviation	1.393
Coefficient of Variation (CV)	1.30%
Median	107.068
5th Percentile	104.895
95th Percentile	109.388

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the unmounted mobile device liquid crystal display (LCD) is 107.096 kg CO₂-eq, which represents a moderate environmental impact for a single display component. This value is primarily driven by the use phase, accounting for 89.6% (95.937 kg CO₂-eq) of the total emissions. The raw material extraction (2.3%), production (2.9%), distribution (1.2%), and end-of-life (4.1%) stages contribute only minor shares, indicating that the display's operational energy consumption during its lifetime is the dominant environmental burden. The top process contributions are listed as unknown, which suggests that the underlying LCA model relies on aggregated or generic datasets rather than specific, traced processes. This limits the ability to pinpoint exact emission sources, but the use-phase dominance strongly implies that electricity consumption for backlighting and driving circuits is the primary contributor. The uncertainty analysis confirms high reliability, with a coefficient of variation of only 1.30% and a narrow 90% confidence interval (104.895–109.388 kg CO₂-eq), indicating that the total footprint estimate is robust.

2. Statement of Assumptions and Limitations

This LCA is based on several key assumptions. The use phase assumes a typical operational lifespan of 3–5 years for a mobile device display, with daily usage patterns aligned with average consumer behavior (e.g., 4–6 hours of active use per day). The electricity grid mix is assumed to reflect the global average, which may not capture regional variations in carbon intensity. For raw materials and production, the model uses secondary data from generic LCA databases (e.g., ecoinvent), which may not fully represent the specific manufacturing processes or supply chains of the producer. The top process contributions being recorded as “Unknown” highlights a limitation: the model lacks granular process-level detail, preventing identification of specific emission hotspots beyond the lifecycle stage level. System boundaries are cradle-to-grave, covering raw material extraction, manufacturing, distribution, use, and end-of-life. However, capital goods (e.g., production machinery), maintenance, and disposal of auxiliary materials are excluded. These restrictions mean the results are indicative but not exhaustive, and improvements in data quality are needed for deeper insights.

3. Suggestions for Improvement

- ****Target the use phase****: Implement energy-efficient backlighting technologies (e.g., mini-LED or micro-LED) and adaptive brightness algorithms to reduce electricity consumption during operation. Encourage users to enable power-saving modes and reduce screen-on time.
- ****Extend product lifetime****: Design displays for easier repairability and modular replacement, and promote refurbishment programs to delay end-of-life disposal. Longer use reduces the per-year carbon footprint.
- ****Shift to renewable energy****: For the use phase, advocate for the use of renewable electricity (solar, wind) by consumers or device chargers with higher efficiency. For production, manufacturers should source renewable energy for LCD fabrication plants.
- ****Improve end-of-life recycling****: Establish take-back programs for LCD panels to recover indium, glass, and liquid crystal materials. Improve separation technologies to increase recycling rates and reduce landfill emissions.

- ****Enhance data quality****: Conduct primary data collection for key production processes (e.g., backlight assembly, polarizer lamination) to replace “Unknown” process contributions. Use region-specific grid mixes and realistic use scenarios to refine the LCA model.